

Weekly Progress Report

Model Development Track: Class Imbalance Research & Legacy
Audits

Jacky Lan

| The 2.72% Challenge

Our current dataset exhibits a severe positive class imbalance, with human presence constituting only **2.72%** of the data.

- **Standard SMOTE is failing:** In bioacoustics, human noises and background wind/rain share highly overlapping spectral features.
- Blindly generating synthetic data creates artificial "positive" points deep within the "negative" noise territory.
- This confuses the model's decision boundaries, driving up False Positives.



Negative Class: 97.28%
(Background Noise / Weather)

Positive Class: 2.72%
(Human Presence)

| Advanced Synthetic Oversampling

To respect the decision boundary of overlapping acoustic classes, we are implementing two boundary-focused algorithms applied exclusively to the training set to prevent test leakage:



1. Borderline-SMOTE

Avoids creating noise by only generating synthetic samples from the "Danger" group—minority points positioned directly on the decision boundary between human and background classes.

Rationale

Based on Han et al. (2005), targeting the decision boundary prevents the dilution of the majority class, which is critical for highly overlapping datasets.



2. SMOTE + Tomek Links

A hybrid approach that generates synthetic data via SMOTE, then actively hunts for overlapping ambiguous pairs (Tomek Links) and deletes them to mathematically "clean" the boundary.

Rationale

Batista et al. / He & Garcia (2009) demonstrate that combining oversampling with undersampling cleanup yields the highest precision-recall stability in noisy sensor data.

Oversampling Results: Borderline-SMOTE

Analysis of decision boundary stability and confusion matrix performance on the acoustic training set.

Training and Evaluating: MLP NN (Borderline-SMOTE)				
ROC-AUC: 0.9958				
Best threshold: 0.80, F1: 0.791				
	precision	recall	f1-score	support
0	1.00	0.99	0.99	91156
1	0.71	0.89	0.79	2342
accuracy			0.99	93498
macro avg	0.85	0.94	0.89	93498
weighted avg	0.99	0.99	0.99	93498

MLP 30 Features ROC-AUC: 0.9960				
Best threshold: 0.30, F1: 0.789				
	precision	recall	f1-score	support
0	1.00	0.99	0.99	91156
1	0.73	0.86	0.79	2342
accuracy			0.99	93498
macro avg	0.86	0.93	0.89	93498
weighted avg	0.99	0.99	0.99	93498

Though the f1-score has not changed between the two iterations, there is a notable increase in recall from a relatively minor decrease in precision. This also demonstrates the tradeoff nature of oversampling: with a higher positive rate comes a greater sensitivity to positive data, but also introduces more false positives. This can also be due to the technique: Borderline-SMOTE works near the positive-negative boundary, so it would be less effective if positive and negative data was intertwined.

| Legacy Model Audit

As part of standardizing our ML Engineering pipelines, I initiated a code audit of previous high-performing models to extract effective hyperparameters and engineered features.

Roadblock: Missing Documentation



I am currently unable to locate Patrick's original Google Colab notebook for the XGBoosting model that recorded exceptionally high performance. The exact hyperparameter configurations and scaling pipelines for that specific iteration are currently undocumented.

Action Item: We have successfully integrated Patrick's engineered features (Cyclical Time, Volume/Wind Ratio, Eerie Silence, 15s Volume Spikes) into the current master dataframe, but we will need to re-run hyperparameter tuning sweeps from scratch if the original notebook cannot be recovered.

| Actionable Next Steps



1. Complete Oversampling Trials

We have successfully generated and visualized results for the Borderline-SMOTE dataset. The immediate next step is to generate the comparative results for the **SMOTE + Tomek Links** pipeline to determine the champion tabular sampling strategy.

2. Explore Data Augmentation

Once tabular synthesis is evaluated, we will explore Audio-Space Data Augmentation. As outlined by Salamon & Bello (2017), physically altering the raw audio via **pitch shifting, time stretching, and background noise mixing** guarantees mathematically viable feature arrays and often outperforms SMOTE on spectral data.